

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (ES6+), Java, VBA

Data & ML: NumPy, Pandas, scikit-learn, PyTorch, spaCy, NLTK, Hugging Face, Neo4j, PowerBI

Cloud & DevOps: AWS, Terraform, Docker, Jenkins, GitHub Actions, Linux (Ubuntu, Amazon Linux)

Web Frameworks: React, React-Native, Angular, Node.js, Flask, HTML/CSS

Additional Competencies: Adobe (Photoshop, Illustrator, Indesign), Autodesk (AutoCAD, Revit, 3DS Max), Unreal Engine

Certifications: AWS Certified Cloud Practitioner, Neo4j Certified Professional, The Complete 2023 Web Developer Bootcamp

RESEARCH PROJECTS & PUBLICATIONS

Oshidero, D. F. O. & Coley, D. A. (2026). **Talking Carbon: A Lexical Approach to Predictive Carbon Analysis via Machine Learning**. *International Journal of Architectural Engineering and Design Management*, pp. 1–34. doi: 10.1080/17452007.2026.2613773

- Introduced an NLP-driven embodied carbon prediction system using transformer-based entity extraction and Histogram-based Gradient Boosting models to analyse conversational architectural design language.
- Deployed full-stack infra with Terraform on AWS, using Python/Docker and React. Project website: www.talkingcarbon.com.

RELEVANT EXPERIENCE

Software Engineer (Full-Stack, AI/ML)

Intropic — AI-Driven Finance & Investment Analytics SaaS Provider

London, UK

May 2026 – Present

- * Contribute to maintenance and development of *The Refinery* (core infrastructure platform) as part of the Platform team, to provide shared tooling and support scalable engineering workflows.

Software Engineer (Data Systems & Infrastructure)

Robert Bosch GmbH & ETAS Ltd. — Embedded Systems for Automotive Industry

Manchester, UK

Sep. 2024 – May 2026

- * Cut analytics platform costs by ~88% (**from \$5.5k to \$675/month**) by replacing managed AWS OpenSearch/OSIS with a self-managed EC2 OpenSearch cluster using Python-based ingesters and parsers, maintaining horizontal scalability with Kubernetes (EKS).
- * Designed pattern mining algorithms for sequence-based machine learning models to analyse temporal diagnostic logs and predict next-likely vehicle faults, with approximately **60% accuracy**.
- * Architected a production-ready, RESTful NLP interface for Neo4j using transformer intent models and LoRA fine-tuning, enabling non-technical users to query graph database with **above 90% success**.
- * Designed and automated system observability for infrastructure operations with Java, Terraform, and Jenkins, managing Linux-based servers (Amazon Linux, Ubuntu) across production and staging environments at scale by monitoring DNS resolution, host connectivity, and service health metrics to generate diagnostic reports for **25+ stakeholders**.
- * Implemented AWS security controls and audit-ready logging for compliance under **ISO** and **NIST** frameworks, and **GDPR** requirements (least-privilege IAM, KMS encryption, TLS in transit, and CloudTrail/CloudWatch).
- * Re-architected legacy ingestion into a scalable Python XML-to-JSON AWS Lambda pipeline processing **130M+ records/day**, handling high-volume document intake via SFTP/SSH, enabling **over 70% latency reduction** in the pipeline E2E.

Data Engineer & Analyst (BI Infrastructure)

KTC Edibles Ltd. — Supplier & Manufacturer for Food & Oil

Wolverhampton, UK

Jun. 2023 – Sep. 2023

- * Engineered an automated database infrastructure by replacing manual CSV workflows with SQL-based ETL pipelines, CRM ingestion scripts, agent scheduling, and validation processes — eliminating **16+ hours** of weekly manual work and enabling **real-time reporting for 80+ employees**.
- * Enhanced BI capabilities by implementing row-level security in dashboards, VBA-based data entry capabilities with audit trails, and improved data models — seeing a **93% improvement** in reported user satisfaction and reporting efficiency.

EDUCATION

Academic Qualifications

Master of Science, Applied Data Science — University of Buckingham

Graduated with Distinction

Master of Science, Computer Science — University of Bath

Graduated with Distinction

Awarded the Global Leaders Scholarship for academic excellence and outstanding leadership.

(Postgrad.) RQF Level 7, Digital and Technology Solutions — University of Buckingham

Awarded with Merit

Honours Bachelor of Science — University of Bath

Graduated with 2:1

Professional Qualifications

Level 3 Diploma, Networking and Cybersecurity

Awarded with Merit

Papertalk: Change Intelligence SaaS for Building Design | *TypeScript (React), Python (ML Libraries), Docker, PostgreSQL*

- * Architected and launched **Papertalk** as a multi-service platform: React/TypeScript clients, FastAPI microservices, Postgres-backed project state, and independently deployable ML engine containers over a private network API.
- * Implemented a shared project data model (facts, assumptions, sources, confidence) driving before/after brief diffs and assessment orchestration via engines; first engine (TalkingCarbon) exposes NLP $\Rightarrow$ HistGradientBoosting inference via FastAPI (/predict, /predict/text).
- * Built CI/CD and infra for production (Docker/GHCR, Kamal, Terraform, Supabase/Postgres on Hetzner boxes) to productize peer-reviewed research into SaaS platform.

Deep RL Models for Increasingly Complex Game Environments: Doom & CartPole | *Python, Pytorch*

- * Developed and implemented deep reinforcement learning algorithms using **Markov Decision Process (MDP)** frameworks and **neural networks (DDDQN, DRQN, PPO, REINFORCE)** for model training on agents in both simple (CartPole) and complex (VizDoom “Defend the Center”) environments.
- * Evaluated agent performance across environments using **experience replay, target networks, reward shaping, and exploration vs exploitation** strategies, leveraging **convolutional feature extractors** on raw visual input to learn navigation, resource management, and survival behaviors in high-dimensional state/action spaces.

Bloom: Gamified Plant Learning Application | *JavaScript, React, React-Native*

- * Led the design and delivery of **Bloom**, using agile development best practices to build a cross-platform mobile app that gamifies plant care education through interactive learning and habit-forming mechanics.
- * Deployed to **12+ physical iOS/Android devices** via Expo for real-world testing, enabling structured **user-interaction data collection** for behavioral analysis and iterative development optimisation.
- * Ran A/B tests and weekly feedback loops on onboarding and learning flows, analysing behaviour signals to drive feature development, achieving **80% increase in satisfaction scores** and measurable gains in user engagement.

Simulation of Relational Database Systems (CRUD & Data APIs) | *Python (Tkinter, SQLite3), SQL*

- * Built a Python Tkinter airline database application backed by SQLite3, including a designed relational schema and a modular data-access layer exposing CRUD APIs for core entities (flights, employees, airports).
- * Implemented queries to database API with input validation, transaction-safe writes, and exception handling to ensure consistent real-time updates.